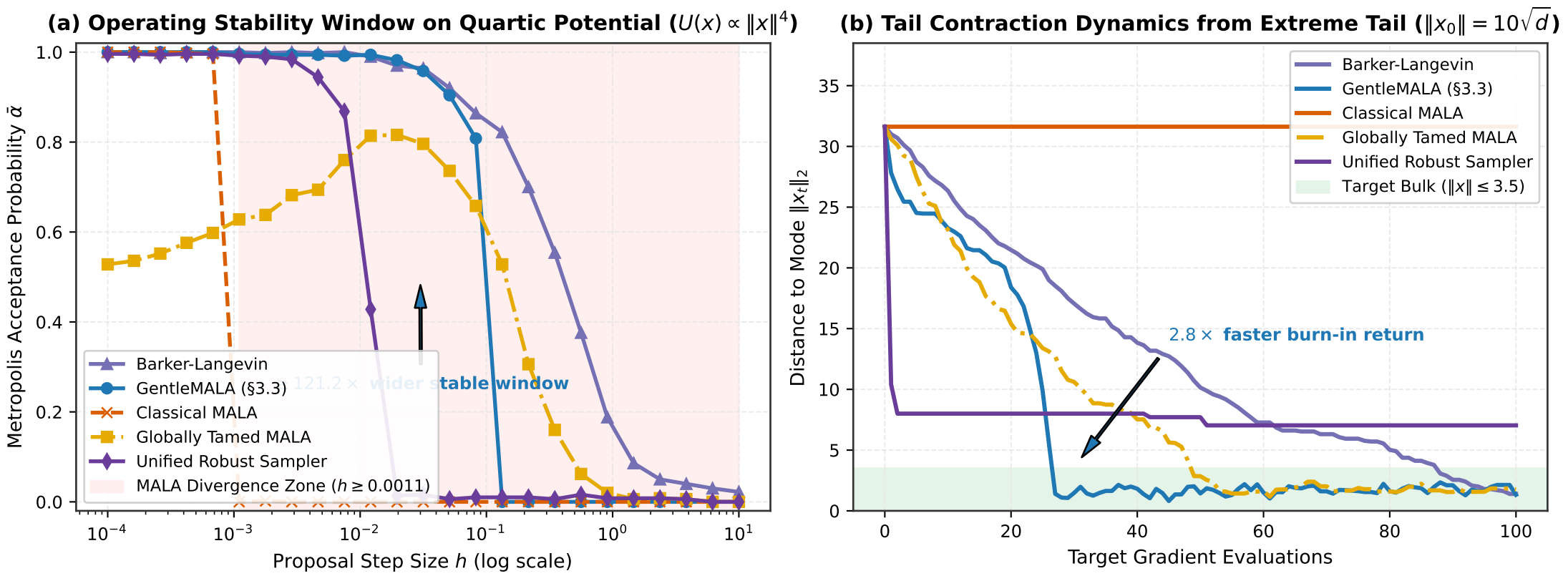
(a) Operating Stability Window on Quartic Potential ($U(x) \propto \|x\|^4$)
Metropolis Acceptance Probability $\bar{\alpha}$
Proposal Step Size h (log scale)
Barker-Langevin
GentleMALA (§3.3)
Classical MALA
Globally Tamed MALA
Unified Robust Sampler
MALA Divergence Zone ($h \geq 0.0011$)
121.2× wider stable window
(b) Tail Contraction Dynamics from Extreme Tail ($\|x_0\| = 10\sqrt{d}$)
Distance to Mode $\|x_t\|_2$
Target Gradient Evaluations
Barker-Langevin
GentleMALA (§3.3)
Classical MALA
Globally Tamed MALA
Unified Robust Sampler
Target Bulk ($\|x\| \leq 3.5$)
2.8× faster burn-in return